

AI Chat Integration

Estimated Monthly Cost

Prepared 2026-06-16 - Model: claude-haiku-4-5

This estimate is grounded in the actual chat implementation (src/chat/chat.service.ts) and current Claude API pricing. The integration runs server-side via the Anthropic SDK tool runner, exposing four read-only analytics tools to the model.

What drives the cost

Factor	Value (from code)
Model	claude-haiku-4-5 (default; overridable via ANTHROPIC_MODEL)
Haiku 4.5 pricing	\$1.00 / 1M input tokens - \$5.00 / 1M output tokens
Fixed prefix per request	System prompt + tool schemas, re-sent on every request
Prompt caching	NOT enabled - prefix billed at full price every call
Tool loop	Tool runner re-sends the full prefix on each loop iteration
History	Stateless backend - frontend resends full conversation each turn

Fixed prefix size is approximately 3,700 input tokens on every request:

- assistant-context.md = 10.4 KB ~ 2,800 tokens
- BASE_PROMPT + current-user block ~ 300 tokens
- 4 tool schemas ~ 700 tokens

Per-message cost

Message type	Input tok	Output	Cost
Simple question, no tool call	~3,700	~250	~\$0.005
Question needing 1 tool call (2 round-trips)	~7,700	~330	~\$0.010

Most questions in this app (token balance, requests, analytics) trigger a tool call, so a blended ~\$0.008-0.01 per user message is a safe planning figure - slightly higher in long conversations as history regrows.

Estimated monthly cost by volume

Assuming ~20 working days/month at ~\$0.01 per message:

Usage scenario	Messages / mo	Est. monthly cost
Light (50 active users x 2 msg/day)	~2,000	~\$20
Moderate (60 active x 5 msg/day)	~6,000	~\$60
Heavy (150 active x 7 msg/day)	~20,000	~\$200
Very heavy	~50,000	~\$500

For an internal coaching tool, the realistic range is approximately \$30-150 per month. Haiku 4.5 is the cheapest current model, so this workload is already cost-favorable.

Two levers worth knowing

1. Prompt caching is the biggest opportunity

The ~3,700-token prefix is identical on every request and every tool-loop iteration, but is currently billed at full rate. Adding `cache_control` to the system block makes repeat hits cost ~0.1x - cutting input cost on the dominant component by up to ~90% for active conversations. Small, low-risk change; highest-leverage optimization.

2. Model override multiplies cost

If ANTHROPIC_MODEL is set, all figures scale directly: Sonnet 4.6 ~ 3x, Opus 4.8 ~ 5x. Keep this workload on Haiku unless answer quality demands otherwise.

Estimates are planning figures based on token approximations (~4 chars/token) and list pricing at time of writing. Actual cost depends on real usage, conversation length, and tool-call frequency. Monitor usage.input_tokens / output_tokens in API responses to refine.